

Using the COSMIC Method to Estimate Agile User Stories

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ABSTRACT

Agile Project Management (APM) is widely used in different software projects from different application domains. APM includes a series of commonly used project management approaches with the intent to handle better uncertainty and unpredictability, which is not always successful. For instance, in a large portion of software projects, accurate planning (and estimating) of whole project lifetime with Agile is difficult. Since a continuously change in (product) requirements occurs as well their incompleteness at the project initialization phase, project plan must be under control and continuously be revised according to its needs and resources. In Agile projects, planning is mainly based on guess estimate of the effort trying to balance the product and the resources not showing the part of each one. In order to improve the guess estimate, this paper proposes an approach moving from the COCOMO (Constructive Cost Model) worked, using the COSMIC measurement method at the micro-level (User Stories) jointly with the quality of the documentation for deploying the functional analysis. The proposed procedure shows that this approach can help the planner to know better why the global effort changes by the time.

Categories and Subject Descriptors

D.2.8 [Metrics]: Product Metrics; D.2.9 [Management]: Cost Estimation; K.6.1 [Project and People Management]: Strategic Information Systems Planning

General Terms

Management, Measurement, Documentation, Standardization.

Keywords

Agile, Planning, Software Project Management, COSMIC, Function Point.

1. INTRODUCTION

In Agile software projects, project plans are not necessarily reliable, because many factors should be considered when planning. Expert Judgment (based on guess estimates) as defined by many authors, including Boehm [1], does not consider separately but globally the change in the nature and overall size of a software product, the resources that can be used during the projects and the estimation of the total effort. From such perspective, most of the time, Agile software projects are only guess estimated [2][3]. Boehm's parametric approach (COCOMO) [1][4] takes into account lines of code (LOC), a series of productivity factors (resources) and a PDR (hours per KLOC, Project Delivery Rate). However it is also possible to use

a Functional Size Method (FSM) instead of KLOC. Functionally sizing the software with a FSM method would be also helpful to prepare scheduling plans, budget plans and resource plans, overcoming the problems and bottlenecks using LOCs (as discussed with the 'productivity paradox') [5].

In most known Agile approaches, User Stories (US) represent a high-level requirement that typically express the functional size (FUR – Functional User Requirement). A US is a brief statement of intent that describes the user demand [20]. The Functional size of a US can be measured using a FSM method, such as COSMIC [6]. The purpose of the COSMIC methodology is to provide a standardized method for measuring the functional size of a software product, no matter to which functional domain it belongs [1]. One of the advantages when adopting a FSM method is that it is technology-independent and moves yet from the high-level requirements produced early in a project. Thus, the technology will not entirely affect the measurement results and planning decisions. COSMIC measurement method is proposed because COSMIC can measure small pieces of functionality and can help to determine if the information is sufficient or not. Applying COSMIC is a good way to find the quality of the documentation [7]. This particular topic about the quality of the documentation is not discussed in this article because of a lack of space.

After measuring US with COSMIC it is possible to determine the effort for each US by: using an internal Project Historical Database [8] or referring to an external repository such as the ones by the International Software Benchmarking Standards Group (ISBSG) [9] or by guess estimations from the project team.

The paper proceeds as follows: Section 2 introduces Agile methods and ISBSG repository. Section 3 presents the proposed parametric approach and its possible benefits with some examples. Section 4 summarizes the main issues for this paper and draws possible future research topics and evolution for this work.

2. PRELIMINARY INVESTIGATION

2.1 Agile Methodology

Agile methodologies are commonly used for better handling uncertainty and unpredictability in the management of requirements during the project lifetime. They are widely used in different project types and very different application domains. The term Agile denotes "*the quality of being agile, readiness for motion, nimbleness, activity, dexterity in motion*" [10] and the management of a software project should also benefit from this quickness. The formal origins for the Agile Alliance moved from the writing of the 'Agile Manifesto' [11] in 2001, proposing a

series of values and principles typically to this new, different way to manage a project. Here the four original values:

- Individuals and interactions *over* processes and tools
- Working software *over* comprehensive documentation
- Customer collaboration *over* contract negotiation
- Responding to change *over* following a plan

Project plans in Agile contexts are prepared according to requirements taken from the users, but involving more actively also the customer during the whole project lifetime. Such requirements are typically express by short sentences (User Stories) expressing the (functional) to-do from the user's viewpoint. Formally, a US is a brief statement of intent that describes something the system needs to do for the user [16]. Such US are generally written on small story cards and they are used during the project implementation for determining the needs of the users.

As example, one of the technique used to determine the amount of work needed to implement the given user story is called Planning Poker. Planning Poker is a Delphi approach that works in the following way:

- individual stories are presented for estimation.
- after a period of discussion, each participant chooses from his own deck the numbered card that represents his estimate of how much work is involved in the story under discussion.
- all estimates are kept private until each participant has chosen a card.
- finally, all estimates are revealed and discussion can begin again [17].

The group, making the discussions for Planning Poker approach, includes all the developers. In Planning Poker approach, it is very common that the estimates differ from one estimator to another; however it is not considered as bad news because, in this case, the high and low estimators will be able to explain their own ideas about the stories and they need to justify their explanations with solid ideas so that they will be able to convince other estimators. So, the discussions begin once more.

The positive side of Planning Poker approach is that many experts come together and each of them shares different ideas with different perspectives. So, the meetings will be really helpful to think on different conditions that may occur during the implementation of the stories. Second advantage is the face-to-face discussions; each estimator needs to convince others about his/her own estimate. This will yield more accurate estimates. Finally, this process is not annoying since it is just like a game.

However, Planning Poker is still not very reliable as long it is just guess estimation. It does not matter whether one person is estimating or a group of people are estimating. It still does not rely on other analysis like historical measures or analogy. The parametric approach prevents those problems using an 'historical repository'.

2.2 The ISBSG repository

The ISBSG D&E (Development & Enhancement) repository is a collection of more than 5000 projects, each one presenting 100+ fields, among them the following information¹:

- functional size (expressed by any of the Functional Size Measurement method and not only, many projects are also sized against LOC)
- Effort in hours (normally this information is related to the development or enhancement part of the life cycle of a project);
- Different characteristics for each project (there are more than a hundred like the domain, the language, the experience of the IT people, etc.).

How to estimate the unit effort of a project using the ISBSG historical database? There are different ways to do it. A typical way consists of such steps:

- Establish criteria to choose typical projects from the historical database (example: MIS domain, java script language, projects between 100 and 800 function points, etc.);
- Find the projects with those criteria in the historical database;
- Calculate the lower, average and higher unit effort (hours per function point);
- Calculate the R square to determine the reliability of the results (more than 70% R square is quite good, less than 50% is not very reliable);
- Decide the unit effort to apply (this could be a range of unit effort. For example, between 5 and 10 hours per function point with an average of 7 hours per function point)

Even with the parametric approach there is still some guess estimate. However when using ISBSG repository there are some facts behind and it is possible to be aware of the reliability of the judgment. Section 3 proposes different steps to implement the parametric approach with COSMIC measurement method and gives some examples.

3. PRESENTATION OF THE APPROACH USED TO IMPROVE PLANNING

This section present the different steps to implement the parametric approach to measure [18] the functional size of US using COSMIC measurement method as part of the process to determine the effort needed to implement each of them.

The purpose of the COSMIC method is to provide a standardized method of measuring the functional size of software whatever its application domain. The COSMIC method was also accepted by ISO/IEC JTC1/SC7 as an International Standard (ISO/IEC 19761) [18][19]. As for the other FSM methods, COSMIC can be applied also from early phases, needing at least high-level requirements to be refined during the project life cycle. In the measurement process data movements are counted which can be defined as "base functional components which moves a single data group type". There are 4 sub-types of data movements: Entry (E), eXit (X), Read (R) and Write (W). The summation of data movements within the established measurement scope returns the number of COSMIC Function Points (CFP).

After using COSMIC to quantify the functional size of each US, the size will be used to obtain the effort in hours. Steps required for this approach are shown in Figure 1.

¹ See www.isbsg.org for more info on the structure of the repository.

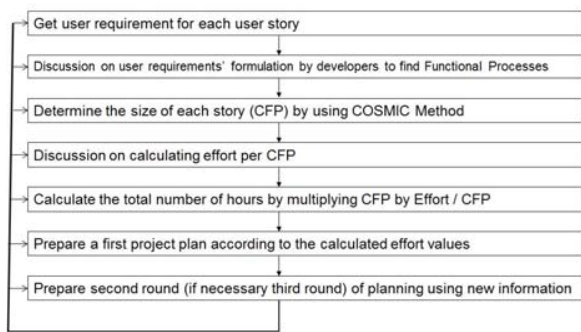


Figure 1. Procedure to obtain effort in hours for a story

From Figure 1, it is possible to see a loop after last step, which means the process continues starting again from any of the others steps. This is needed because within the first (or previous) iteration some information may be missing. Then, it is needed to catch the missing information during next step and the process needs to be iterated until all information will be collected and final values are calculated. Next sub-sections will provide details for all the 7 steps.

3.1 Getting User Requirements in Terms of User Stories

In the first step of the approach, User Requirements are taken in the form of US. As already mentioned, a US is a brief statement of intent that describes something the system needs to do for the user [16]. A US describes a feature of the software [20]. Each story is formulated in one or two sentences in the language of the customer, and typically written on an index card. Rather than conveying all details about the feature, details that are likely to change during the project, the cards serve as reminders for conversations to be had about the features. The details are then fleshed out in the conversations, and conveyed and documented in the form of tests.

First step is giving only the stories at a high level. The discussion is oriented not on the effort but on the job that should be done with the information provided to know the functional size of the story. First US can be read 'Customers can rent movies on-site'. Instead of discussion how long it takes to do this job, Figure 2 summarizes the discussion and main inputs and elements for an estimate with COSMIC:

*What information is needed to rent a movie?
Assuming there is a need for basic information on the client (name, address, telephone, client number) plus information on the film plus information on price, then, there are at least 3 data groups. According to COSMIC rules there are 3 Entries, 3 Reads, 3 Writes and 4 eXits including the error message. PDR is assumed to be c.a. 5 hours/CFP for this type of project.*

Figure 2 – Analysis and assumptions for the US 'Customer can rent movies on-site' for a COSMIC functional count and determination of related effort

From those questions (Figure 2), it is possible to focus on the information given by the user.

3.2 Formulations and Identification of Functional Processes

After user stories are taken from the users, the developers, in terms of technical implementation, will formulate them and then the functional processes of those user stories will be determined. Identification of the functional process is at the mapping phase of COSMIC measurement method (Table 1).

In order to be measured, a functional process needs to be identified. A functional process is an elementary component of a set of Functional User Requirements (FUR) comprising a unique, cohesive and independently executable set of data movements. It is triggered by a data movement (an Entry) from a functional user that informs the piece of software that the functional user has identified a triggering event. It is complete when it has executed all that is required to be done in response to the triggering event [18][6].

Table 1. Procedure Steps to apply COSMIC method – Pt. 1[6]

MEASUREMENT STRATEGY (MS)	MS.01 - Definition of purpose of the measurement
	MS.02 - Definition of scope of the measurement
	MS.03 - Identification of functional users
	MS.04 - Identification of level of granularity
MAPPING (MA)	MA.01 - Application of Generic Software Model
	MA.02 - Identification of functional processes
	MA.03 - Identification of objects of interest and data groups

From Figure 2 'Customer can rent movies on-site' is identified as a functional process. Behind this brief statement a number of questions arise as also suggested in Figure 2. The measurer will try to answer those questions before determining the functional size of each US. By following COSMIC measurement method rules it is not difficult to raise some questions about the available information²:

- how many functional processes within a story
- is the story imply a change in the database (write)
- what is the trigger of the functional process
- etc.

As the project will go on, and more information is coming in, there is a possibility to have a number of changes in the software product. Next step is the determination of the functional size of each US n terms of CFPs.

3.3 Determination of the Functional Size of Each US

After USs are defined, their functional size need to be calculated. Measurement of the stories is done by using the COSMIC method, which gives the results in terms of the size of each functional process (COSMIC Function Points or CFP) from the information available at that time. This completes the Mapping phase (table 1). Remaining steps to apply COSMIC method are shown in Table 2.

² The criteria to have a good quality user story in general is a larger discussion. The aim here is to obtain a number of COSMIC function points and determine if the information is sufficient to have a precise and accurate number.

Table 2. Procedure Steps to apply COSMIC method – Pt. 2 [6]

MEASUREMENT (ME)	ME.01 - Identification of data attributes
	ME.02 - Identification of data movements
	ME.03 - Application of measurement function
	ME.04 - Aggregation of measurement results

Those steps will be explained and applied later. Remember that any FSM method only sizes product FURs: therefore in sub sections 3.5 and 3.6 of this parametric approach, the effort related to NFR (Non-functional requirements) must be taken into account.

Using the functional process mentioned in sub section 3.1 (figure 2) the size of the User Story (US) is 13 CFPs (3 Entry, 3 Read, 3 Write, 4 Exit). This should be done for every functional process using the information available.

Table 3 shows the measurement results of 10 US and 15 functional processes (FP), which means there is a possibility to have more than one functional process for each story. Also because the knowledge about the functional process evolves by the time, the CFPs also change. The first three columns present the number of CFPs per functional process and the last two the absolute difference between the different phases of the project.

Table 3 shows also the evolution of the size of the project at different phases of the project from 83 CFPs to 132 CFPs. There are only 3 phases in this example, but this is not limited. This can be seen as different iterations to obtain the size when preparing the project plan (sub section 3.6).

Table 3. Measurement results for the 10 US and 15 FPs

	CFP for Phase 1	CFP for Phase 2	CFP for Phase 3	Change for Phase 1 - 2	Change for Phase 2 - 3
US 1	13	13	12	0	1
US 1a	-	17	16	17	1
US 2	10	4	4	6	0
US 2a	-	-	10	0	10
US 3	4	4	4	0	0
US 3a	-	6	6	6	0
US 4	12	13	12	1	1
US 5	8	8	8	0	0
US 5a	-	-	10	0	10
US 6	12	11	10	1	1
US 7	8	8	8	0	0
US 7a	-	11	10	11	1
US 8	4	6	6	2	0
US 9	8	9	9	1	0
US 10	4	7	7	3	0
Total	83	113	132	48	25

3.4 Discussion on Calculating Effort per COSMIC Function Point

After the functional size for each US is determined, it is time to calculate the needed effort for implementing that US. In order to do this, effort per CFP value needs to be determined. There are different ways for determination of this value. Group of experienced people may find an agreement on such value or derive it from a repository of previously implemented projects, selecting their more feasible PDR (effort/CFP) range of values after applying some filters by the needed project characteristics (e.g. development type, programming language(s), application domain, etc.). A valuable reference point for benchmarking and/or when the organizational Project Historical Database (PHD) is not present or contains few reference data is give by ISBSG repositories [10]. This step is also important since it will be useful for the calculation of total effort needed. In order to be effective as much as possible in the estimation process, it must be remembered that currently ISBSG data does not split the effort by requirement type (FUR vs NFR), but only propose a split by high-level SLC phases (as discussed in [21]). It typically leads a project manager to need more information and time before deriving the final project plan, scheduling also effort for typical project roles not expressing ‘functional-related effort’, such as the project manager itself, the FSM counter, a QA people, an usability expert or an accounting people dealing with administrative tasks [15]3.

Therefore a suggestion is to derive – at least with a high-level view – the proportion between the effort derived from FUR and NFR for each project type, in order to properly manage PDR variability and ranges in the estimation process [8].

For the example of the Video Solution, could determine the Effort per CFP value at 5 hours (see sub section 2.2). Thus, the total effort for the first round of the first US was 65 person-hours (13 CFP * 5m-hrs/CFP).

3.5 Calculation of the Total Number of Hours

After determining the typical PDRs for a project according to its characteristics and constraints, next step is to apply those PDRs to each US and therefore summing them within the boundary of the overall project4. This is one of the easiest steps since this operation is quite mechanic, just referring to multiplications and summations. PDR values will be multiplied by the number of measured CFP for each US, and such value will return the needed effort for that US. Then in order to calculate the total effort of the project in terms of person-hours (p/hrs), the sum of such calculated values will be taken. Here is an example for this calculation and the generic formula:

$$\begin{aligned}
 \text{User Story A} &= 5 \text{ CFP} \\
 \text{Effort per CFP} &= 2 \text{ p-hrs/CFP} \\
 \text{Effort for User Story A} &= 5 \text{ CFP} * 2 \text{ Hours/CFP} = 10 \text{ p-hrs}
 \end{aligned}$$

³ Such kind of PDR (or productivity) including all the project/US effort can be called ‘nominal’ and not ‘strictly functional’, because including also NFR-related effort, not directly deferrable to the measured CFP [15].

⁴ Note that different PDRs could be applied to the different US because of different technological and environmental combinations included in the project scope under measurement.

$$\begin{array}{ccccc} \text{Overall} & \text{Project} & \text{Effort} & \text{(person-hours)} & = \\ \sum_{i=1}^n CFP_{USi} * E_{USi} & & & & \\ [(CFP_{US1} * E_{US1}) + (CFP_{US2} * E_{US2}) + \dots + (CFP_{USn} * E_{USn})] & & & & \end{array}$$

where:

n is the number of the story,

CFP_{USn} is the number of CFP for n -th US,

E_{USn} is the PDR (effort per CFP) for the n -th US.

3.6 Preparation of Initial Version of the Project Plan

Next step is the preparation of the project plan. This step is under the responsibility of the project manager (or e.g. the Scrum Master, in SCRUM terms). Since, he/she knows the total number of hours, he/she can prepare the first versions of schedule plans, resource plans or budget plans. Project manager will split the project into different iterations (e.g. the SCRUM's Sprints) according to the overall functional size, effort and duration of the project moving from the requirements and information at such time by the customer. For instance, one possible criteria for splitting US by iteration is their rating using the INVEST criteria [20]⁵. As said before in step 3.5, it is critical to know and have (possibly more and more during time) the knowledge about the balancing between FUR and NFR-related effort for a proper estimation and schedule of the different project team roles.

A more measurement-based project plan will return a better reliability and accuracy, providing benefits for all the stakeholders.

3.7 Preparation of the Second Phase for Planning

After initial version of the project plan is ready, a second phase starts according to more precise user requirements. Since in the second round more details will be owned in terms of user stories, the measurement results may change in this second phase. Some uncertainties and incomplete parts of the stories will have been clarified in this second phase. After applying another measurement process; the project plan may be updated according to the new values. This is not restricted with just those cases; there may be additional phases according to the needs.

This approach brings the advantages of, not only guess estimating the effort values, but having a systematic way of measurement of the product and then determining the needed efforts. Additionally, this will help organizations to have a trustable and repeatable consistent way for project planning. By using this approach in many different projects, organizations will have more consistent project plans.

4. CONCLUSIONS AND FUTURE WORKS

It is commonly accepted and thought that management of Agile projects has great amount of uncertainty and unpredictability. In order to solve those problems, different approaches have been developed, such as Planning Poker. Such approaches are useful,

however both just guess estimate project effort and project plans would still depend just on expert judgment. As a result of this, project plans are not as reliable and applicable when approaches used are not based on facts.

In this paper, a different approach was proposed in order to overcome such difficulties. Instead of guess estimation, a FSM method was used for determining the product functional size for the User Stories (US) determined after eliciting requirements from stakeholders. The product functional size derived is measured using the COSMIC method in COSMIC Function Points (CFP). As a result of such measurement, needed effort for the project will be calculated more accurately. So, effort plans, budget plans and resource plans of the projects will be more reliable with this approach.

Additionally, there is still a little amount of guess estimation in determination of the effort per CFP for some US. In fact, this guess estimate does not represent the effort level for all the US in a project, but just for one or more of them. Even if this guess estimation has very little place in the approach, it may be a good work if this guess estimation would be completely overcome.

Finally, this approach was presented in the context of a small project. There is a need to experiment it in the context of a big organization with multiple projects. Measuring each project at different phases, the organization will probably cope better with the risk having multiples projects with increasing and non-controllable size. But at the same time, gathering all those data into an internal Project Historical Database (PHD), keeping ideas from the ISBSG repositories, will give organizations a further opportunity to have at their disposal more and more quantitative inputs for refining estimates and reduce as much as possible the 'cone of uncertainty' [1], saving time and money.

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